CIS 515 AI and Data Analytics Strategy

Assignment 2

Vision Transformer (ViT)

- *All assignments are **individual effort with group consultation**. This means you are allowed to brainstorm solutions with your assigned (project) groups, consult with them on your approach, help each other debug, and clarify programming and deep learning concepts with them. However, you are **not allowed** **to copy code** from one-another. You are required to work on your own assignments and turn in **individual submissions**.*
- *Refer to Syllabus for the late submission policies.*
- *Recall that you have signed the Honor Code. Penalties for academic integrity violations can range from a letter reduction in final course grade (at a minimum) to expulsion from the program and School of Business.*

DESCRIPTION

Assignment 2 leverages the lecture topics of pre-trained models and transfer learning for image classification. In this assignment, we transition from CNN architecture to transformer-based architecture. While we used Tensorflow for Assignment 1, in this Assignment we will use with PyTorch deep learning framework as Pytorch is now primary learning framework for transformer library as of 2025.

TASKS [15 POINTS]

SETUP

Load a starter notebook in Colab. Use a run-time with GPU.

On you Google drive create a folder for data:

`/content/drive/MyDrive/CIS515/competition_data/public`

- Save train/val/ test_public.csv files in this location.
- Create a new session at your LLM of choice and ask all the questions about this assignment in this session only. You will be asked to export and submit your session together with a notebook. The goal of this submission is to understand general.

- Add code in the starter notebook
- Write a text answer in the Assignment 2 framework Word doc

Submission Instructions:

- You must submit THREE files:
- 1. <LastName>Assignment2.ipynb — Your completed Jupyter notebook with all code cells run
- 2. <LastName>Assignment2_Answers.docx — This completed answer sheet with all written responses and screenshots
- 3. <LastName>Assignment2.pdf — Your exported LLM session log used during the assignment

Note: All code stays in the notebook. This document is for written answers and screenshots only.

Part 1 Vision Transformer Setup [1 point]

Instructions: *Follow instructions in the Google Colab Pro setup to enable advanced GPU options. In this assignment, we apply a pre-trained Vision Transformer (ViT) model (vit-base-patch16-224) to make predictions on the Oxford Flowers dataset (dpdl-benchmark/oxford_flowers102).*

Question 1.1 [0.5 point] Research the vit-base-patch16-224 model. What datasets was it pre-trained on and what classes does it include? Do you expect it to be a good fit for the Oxford Flowers dataset? Why or why not?

Type your answer below:

Question 1.2 [0.5 point] Refer to Lab B for ViT code hints. Run the code up to Load and Preprocess Dataset. Load processor using ViTImageProcessor.from_pretrained function. Run “Load and Preprocess Dataset” and Training code blocks.

Part 2 Baseline Model Performance [2 points]

Instructions: *Run the code up to and including the baseline model in the “Baseline Model Performance” notebook block*

Question 2.1 [1 points] In one sentence, how would you evaluate the baseline model's performance?

Type your answer below:

Question 2.2 [1 points] In your opinion, what is the reason for the observed model's performance?

Type your answer below:

Part 3 Optuna Hyperparameter Search [7 points]

Instructions: *Run the code up to "Run Optuna Study" block.*

Question 3.1 [3 points] Update train_and_evaluate() function with additional hyperparamters and update ranges for current hyperparameters if needed.

Question 3.2 [2 point] For each additional hyperparameter you added to the Optuna objective function, explain in one sentence why you expect it to affect model performance.

Type your answer below. Cover every hyperparameter you added beyond the three already provided (lr, warmup_ratio, scheduler_type).

Question 3.3 [2 point] Run Optuna Study code block. Next run code in "Best Model vs Baseline Model Comparison" to compare the best model found by Optuna vs the baseline model. Describe the differences and explain what you believe drives the improvement. Write one paragraph.

Type your answer below. Reference specific accuracy numbers from your results.

Part 4 Final Model [2 points]

Instructions: *Run Navigate to the "Final Evaluation" code block.*

Question 4.1 [2 points] Use the train_and_evaluate() function to reproduce the best model with the corresponding hyperparameters, create the best model object, and generate the test accuracy.

Part 5 Prediction & Kaggle Competition Submission [3 points]

Instruction*: Navigate to "Predict on the Test set" code block*

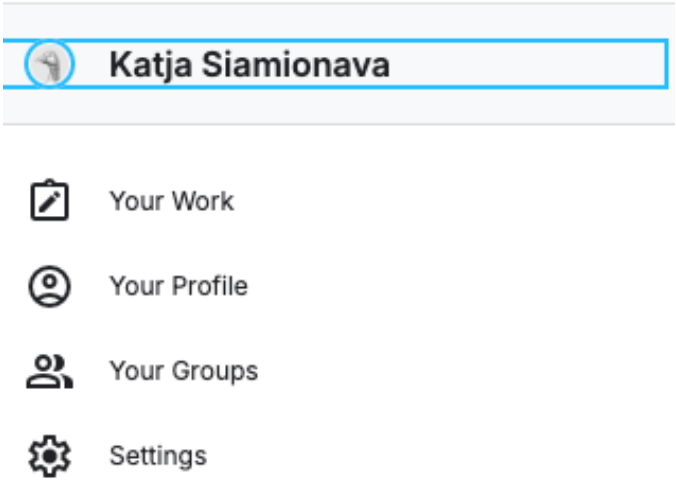
Question 5.1 [0.5 points] Add your ASU ID as STUDENT_NAME.

Question 5.2 [1 point] Make predictions using best_model and save your predictions.

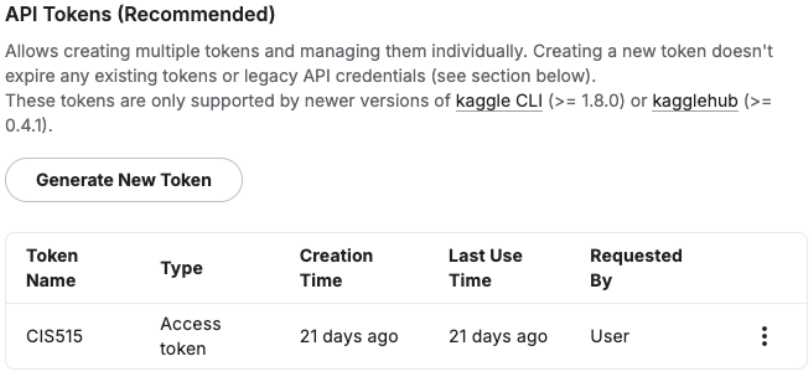
Instructions: *Navigate to the "Kaggle Competition Submission" block.*

Question 5.3 [0.5 points] Create a Kaggle account and generate a Kaggle API token. Add your token and username to the notebook's Secrets tab.

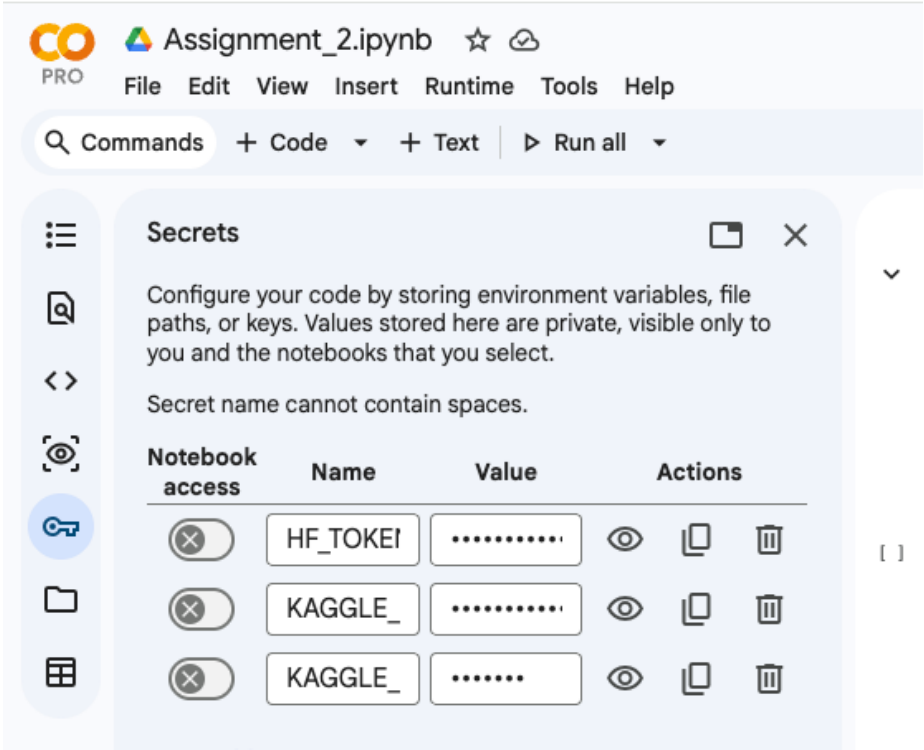
- *Go to https://www.kaggle.com/ and create an account if you do not have one.*
- *Click on the user icon in the top right corner. Select Settings from the side menu.*



- **Scroll to the API options. Create an API Token**



- *Copy the token!*
- *In the Colab Noebook, click on Secrets in the left side menu. Create 2 secrets:*
  a) ***Kaggle user name****. Name it "KAGGLE_USERNAME". Type in your Kaggle user id in the value cell.*
  b) ***Kaggle token****. Name it "KAGGLE_API_TOKEN". Paste your Kaggle token in the value cell.*
  Note: *when you run "Kaggle Competition submission" cell, it will use these secrets as an input.*



Question 5.4 [1 point] Run code for Kaggle submission. Include a screenshot of your successful submission as shown on the Kaggle leaderboard. To access Kaggle

Note: *You may make multiple submissions until you are satisfied with your performance. After reviewing your test results, you can revisit your model, improve it further, and submit again.*

After running the Kaggle submission cell, paste a screenshot confirming your successful submission below.

[Paste successful Kaggle submission screenshot here]

Part 6 Extra Credit [2 points]

If your model scores in the top 10% on the final leaderboard, you will receive 2 points of extra credit.